Vivek Anupindi

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Professional Summary

Ph.D. student in Computer Science at the University of Maryland, Baltimore County, with research interests in machine learning for healthcare, clinical NLP, healthcare analytics, predictive modeling, and intelligent data systems. Experienced researcher and data engineer with 4 years developing data pipelines, machine learning workflows, and cloud-based analytics systems, with current work focused on clinical text classification and temporal health prediction.

Education

University of Maryland, Baltimore County — Ph.D. in Computer Science 2026 – Present
Baltimore, MD
Saint Louis University — M.S. in Information Systems GPA: 4.0/4.0 Aug 2023 – May 2025
St. Louis, MO
B.V. Raju Institute of Technology (JNTU Hyderabad) — B.Tech. in Computer Science GPA: 9.0/10.0

Research Interests

Machine Learning for Healthcare • Clinical NLP • Healthcare Analytics • Predictive Modeling • Time-Series Modeling • Data Engineering • Intelligent Data Systems

Relevant Graduate Coursework

Machine Learning • Artificial Intelligence • Big Data Analytics • Cloud Computing • Research Methods • Health Informatics • Advanced Databases

Publications

- **Vivek Anupindi.** "Comparative Evaluation of Logistic Regression for ICD-10 Code Classification Using the CodiEsp Clinical Text Dataset." *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, vol. 13, no. XII, 2025. DOI: 10.22214/ijraset.2025.76293.
- **Vivek Anupindi.** "Automated ICD-10 Coding from Spanish Clinical Text: A Comparative Study." *Under Review.*
- **Vivek Anupindi.** "Data Engineering Patterns for Cloud-Native Analytics." *Under Review.*

Research Experience

Automated ICD-10 Coding from Spanish Clinical Text — Independent / Published Research

- Developed an NLP pipeline using the CodiEsp clinical text dataset for automated ICD-10 diagnostic code classification.
- Investigated TF-IDF representations and Logistic Regression for multi-label clinical text classification.
- Examined document-label alignment, missing data, and imbalanced diagnostic codes.

AI-Assisted Glucose Prediction for Type 1 Diabetes — Independent Research

- Designed a predictive framework for short-term glucose forecasting using temporal machine learning methods.
- Explored LSTM and transformer-based architectures and features incorporating insulin timing, meals, and physical activity.

Physician Burnout Predictive Analysis — Master's Research Project

- Conducted exploratory and statistical analysis of physician burnout using workload, behavioral, and emotional indicators.
- Investigated relationships among burnout-related variables and specialty-specific trends.

Data Engineering for Cloud Analytics and ML Deployment — Independent Research

- Investigated cloud-native ETL architectures using Snowflake, Apache Airflow, and dbt.
- Examined pipeline optimization, warehouse scaling, and metadata-driven orchestration for downstream machine-learning workloads.

Selected Research Projects

Smart Health Monitoring System — Independent Healthcare Analytics Project

- Built a Python application to track heart rate, blood pressure, oxygen saturation, temperature, and blood glucose measurements.
- Implemented rule-based anomaly flagging and generated basic health reports to support continuous monitoring and early alerts.

Automated Access Management System Integration — SLU ITS — Systems Integration Project

- Integrated the university's iPass (TeamDynamix) ticketing and project-management system with Active Directory to automate user-access workflows.
- Developed secure provisioning logic and workflow rules for user access lifecycles, followed by end-to-end validation of synchronization

between systems.

Research Methods

Clinical text preprocessing • Multi-label classification • Statistical analysis • Experimental evaluation • Data quality assessment • ETL pipeline design

Teaching Experience

Teaching Assistant — CMSC 487: Network Security — University of Maryland, Baltimore County Fall 2026

- Assist with grading and evaluation of course assignments and assessments.
- Support students with questions related to network security concepts and course material.
- Provide feedback on student work and assist with course administration and instructional activities.

Employment

Tata Consultancy Services (TCS) — Data Engineer Consultant May 2025 – Dec 2025
Remote, USA

- Developed Python- and Airflow-based ETL pipelines processing 2M+ daily API records into Snowflake.
- Optimized SQL and dbt transformations, reducing Power BI query times by 35%.
- Implemented automated data-quality validation using Great Expectations.
- Developed configuration-driven DataOps pipeline templates using YAML and Jinja2.

IBM — Data Engineer Jan 2020 – May 2023
Bangalore, India

- Modernized enterprise ETL workflows using Docker, OpenShift, and Cloud Pak for Data.
- Developed data virtualization and multi-source integration solutions across PostgreSQL, Oracle, and NoSQL systems.
- Designed Kubernetes-based workload management for cloud and hybrid ETL environments.
- Developed CI/CD automation using Jenkins, OpenShift Pipelines, and ArgoCD.

Service

- **Computer Science Research Reading Group** — Srinivasa Ramanujan Computer Science Education Undergraduate Research Group. Led reading-group activities focused on computer science research and technical literature.

Honors & Awards

- **Graduate Dean's List**, Saint Louis University.
- **Graduate Merit-Based Scholarship**, Saint Louis University.
- **Academic Excellence**, M.S. in Information Systems, GPA 4.0/4.0.

Presentations & Professional Activities

- DFI–India Engineering Conference, 2022 — Participant.
- Indian National Academy of Engineering (INAE) Events — Participant.

Technical Skills

Programming: Python, SQL, R, Bash
Machine Learning: Scikit-learn, TensorFlow, PyTorch, NLP, Transformers, Time-Series Modeling
Data Engineering: Apache Airflow, dbt, Kafka, DataStage, Azure Data Factory
Cloud & Data Platforms: Snowflake, AWS, Azure, IBM Cloud, Kubernetes, OpenShift
Analytics & Visualization: Power BI, Tableau, Exploratory Data Analysis
Data Quality: Great Expectations, Schema Validation, Data Lineage

Certifications

Microsoft Certified: Power BI Data Analyst · Databricks Certified Data Engineer · Databricks Certified Generative AI · Databricks Fundamentals · Microsoft Azure AI Fundamentals · Microsoft Azure Fundamentals · IBM Cloud Core · IBM Python